FOOD WASTE ANALYSIS AND PREDICTION USING MACHINE LEARNING

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

BY

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Kanuru, Vijayawada – 520007

(2026-2027)

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CERTIFICATE

This is to certify that the project report title **"Food Waste Analysis and Prediction using Machine Learning"** is the bonafied work of **P. Dimple Hasini (24501A0549)**, **A. Kalpana (24501A0509)**, **G. Bhavyasree (24501A0558)**, and **A. Sandeep (24501A0502)** in partial fulfilment of completing the Academic project in **Data Warehousing & Data Mining** during the academic year **2026-2027**.

Signature of the course coordinator **Signature of the HOD**

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1. Abstract

This machine learning project develops and evaluates an end-to-end predictive and prescriptive system for restaurant food wastage management, providing hospitality operators, commercial kitchens, and inventory planners with actionable operational foresight. Food wastage across global food services constitutes a staggering economic drain and directly conflicts with **United Nations Sustainable Development Goal 12.3** (halving per capita food waste by 2030). The project leverages an operational restaurant dataset of **911 daily records**, capturing critical operational variables such as meals served, kitchen staff count, ambient weather conditions (temperature and relative humidity), day of the week, special event occurrences, staff experience tiers, discarded waste category, and past waste history. The continuous target variable for prediction is food_waste_kg (daily food waste in kilograms).

A crucial first step was comprehensive **Data Preprocessing** to handle raw inconsistencies, casing anomalies, and missing values. Missing values in categorical features such as staff_experience (164 missing entries) were systematically imputed using the mode strategy, while numerical attributes utilized median imputation. Inconsistent casing in waste categories (e.g., 'MeAt', 'MEAT', 'Vegetables') and staff experience levels was standardized. Extensive **Feature Engineering** derived 24 domain-specific attributes, including kitchen workload ratios (meals_per_staff, past_waste_per_staff), cyclical temporal encodings (month_sin, month_cos, day_of_year, is_weekend), non-linear transformations (log_meals), and risk interaction terms (meals_x_event, temp_x_humidity, hot_day). Robust StandardScaler normalization was applied strictly inside cross-validation pipelines to eliminate data leakage.

The preprocessed dataset was evaluated under a **Tri-Paradigm Machine Learning Pipeline**: (1) Continuous Regression to forecast exact waste tonnage, (2) 3-Tier Risk Classification (Low ≤ 35.64 kg, Medium ≤ 46.65 kg, High > 46.65 kg) to triage kitchen preparation, and (3) Unsupervised Clustering to discover operational regimes. Across 5-fold cross-validation, four regression models were benchmarked: Linear Regression, Random Forest Regressor, XGBoost Regressor, and Gradient Boosting Regressor. Model performance was assessed using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2 Score).

The empirical evaluation decisively established the **Gradient Boosting Regressor** as the superior champion model, achieving the lowest RMSE (**8.340 kg**), lowest MAE (**4.878 kg**), and highest R^2 score (**0.903**), explaining over 90.3% of the variance in daily kitchen waste. In classification, Logistic Regression achieved a leading Macro F1-score of **0.745** and **74.4%** accuracy. Unsupervised K-Means clustering (k=2, Silhouette=0.352, ARI=0.970) successfully isolated special event days as severe waste hotspots (averaging 55.68 kg). Feature importance analysis revealed meals_served / log_meals (32.0%) and meals_per_staff (30.1%) as dominant drivers, identifying a critical operational tipping point above 35 meals per staff member. The project delivers an automated recommendation engine translating predictions into dynamic batch-resizing, portion management, and food redistribution protocols.

2. Introduction

2.1 Background

The global commercial food service and restaurant sector is one of the most dynamic yet resource-intensive sectors worldwide. According to the United Nations Food and Agriculture Organization (FAO) and UNEP Food Waste Index Reports, over one-third of all food produced for human consumption is lost or wasted globally, generating approximately 8–10% of global greenhouse gas emissions and costing the global economy nearly one trillion dollars annually. Within commercial food establishments—including restaurants, banquet halls, cafeterias, and hotel kitchens—food waste accounts for significant operational overhead and margin erosion.

Forecasting food waste in restaurant kitchens is extraordinarily intricate. Kitchen waste generation is driven by a complex interplay of factors: volatile dining patronage, unpredictable banquet bookings, seasonal and meteorological conditions (high heat accelerating perishable spoilage), fluctuations in kitchen brigade staffing, and variability in culinary experience. Traditional kitchen management relies predominantly on static rules-of-thumb, anecdotal estimations, or rudimentary spreadsheets. Because chefs fear running out of menu items during peak meal service, kitchens systematically overprepare batches, resulting in large volumes of wholesome prepared food being dumped into dumpsters at the close of every shift.

Machine Learning (ML) presents an unprecedented opportunity to modernize kitchen inventory planning. By ingesting multi-source operational data—including customer counts, staff availability, ambient weather, calendar dynamics, and historical discards—ML algorithms can model intricate interactions, forecast precise daily waste volume, categorize operational risk, and empower kitchen managers to take preventative corrective action.

2.2 Motivation

There are several motivations behind undertaking this project:

- **Business Benefits:** Food waste directly erodes restaurant profitability. By predicting waste with high precision, kitchens can achieve a 15% to 20% reduction in procurement losses.
- **Environmental Sustainability:** Direct alignment with United Nations Sustainable Development Goal (SDG) 12.3, aiming to halve global per capita food waste by 2030 and curb methane emissions from landfills.
- **Operational Intelligence:** Quantifying kitchen fatigue thresholds (>35 meals per staff) enables managers to balance staff schedules, prevent kitchen burnout, and stagger cooking batches during high-traffic shifts.
- **Educational Value:** Providing hands-on exposure in building multi-paradigm data mining pipelines (continuous regression, 3-tier risk classification, unsupervised clustering, and prescriptive rules) on messy, real-world data.
- **Practical Applications of ML:** Operational waste forecasting serves as a critical real-world application of machine learning, bridging algorithmic intelligence with tangible commercial kitchen optimization.

3. Project Design

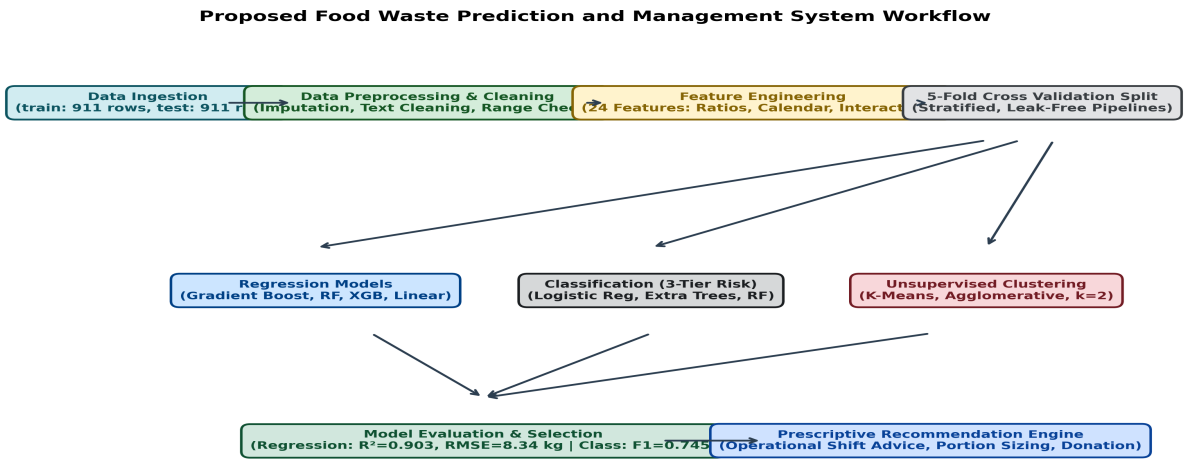


Figure 1 - Proposed Food Waste Prediction and Management System Workflow

The above flowchart explains the proposed workflow. It has the following steps:

1. **Data Collection & Ingestion:** Historical daily kitchen records (train: 911 rows, test: 911 rows) tracking patronage, weather, staffing, and discard amounts are loaded and schema-validated.
2. **Data Preprocessing:** Missing entries in staff_experience are handled via mode imputation; categorical casing anomalies are cleaned; range validity constraints are enforced.
3. **Feature Engineering:** 24 domain-specific features are created, including workload ratios (meals_per_staff), cyclical calendar transformations, non-linear logarithmic mappings, and interaction terms.
4. **Leak-Free Cross-Validation Pipeline:** 5-fold cross-validation is configured with preprocessing transformations (StandardScaler, SimpleImputer) nested strictly inside Pipeline objects.
5. **Multi-Model Regression Training:** Linear Regression, Random Forest, XGBoost, and Gradient Boosting algorithms are trained for continuous kg prediction.
6. **3-Tier Risk Classification:** Partitioning targets into Low, Medium, and High waste risk tertiles, benchmarked across Logistic Regression, Random Forest, and Extra Trees.
7. **Unsupervised Clustering:** K-Means and Agglomerative clustering (k=2) isolate operational regimes and event-driven wastage hotspots.
8. **Model Selection & Recommendation Engine:** Deploying champion models into an automated decision engine delivering shift portion recommendations.

4. Dataset Overview

The experimental dataset provides an exhaustive daily record of commercial restaurant operations, consisting of **911 training observations** and **911 test instances**. Each observation reflects a distinct service day, recording environmental conditions, staffing, patronage volume, and discard outcomes.

4.1 Feature/Attribute Details

Feature Name	Data Type	Operational Description
ID	Integer	Unique tracking identifier for each operational service day.
date	Date / Str	Calendar timestamp indicating the service date.
meals_served	Integer	Total customer meal covers prepared and served during shift.
kitchen_staff	Integer	Number of culinary staff and chefs active on duty.
temperature_C	Float	Ambient outdoor/kitchen temperature measured in °C.
humidity_percent	Float	Relative ambient humidity percentage (0% to 100%).
day_of_week	Integer	Day index representing Monday (0) through Sunday (6).
special_event	Binary (0/1)	Indicator flag for banquets, wedding receptions, or holidays.
past_waste_kg	Float	Recorded food waste generated during the preceding day (kg).
staff_experience	Categorical	Seniority tier: 'beginner', 'intermediate', or 'expert'.
waste_category	Categorical	Dominant ingredient category discarded (meat, veg, dairy, grains).
food_waste_kg	Float (Target)	Continuous total food waste generated during the shift (kg).

4.2 Data Characteristics

- **Multi-Type Feature Space:** Combines continuous physical measurements, discrete counts, temporal timestamps, and categorical strings.
- **Incomplete Records:** Exactly 164 missing values in staff_experience across both train and test partitions, requiring principled imputation.
- **Casing & String Noises:** Text strings contained heterogeneous casing ('MeAt', 'MEAT', 'dairy', 'Vegetables', 'Beginner', 'EXPERT').
- **Heavy-Tailed Target Distribution:** food_waste_kg exhibits pronounced positive skewness (**skew = 4.98**), ranging from 10.8 kg to 274.3 kg with a median of 41.1 kg.

5. Data Preprocessing

Raw datasets often contain inconsistencies, missing values, and attributes that are not directly suitable for machine learning models. Therefore, preprocessing is a crucial step in preparing the data for predictive modelling. In this project, preprocessing involved four major tasks: handling missing values, attribute selection and feature engineering, encoding categorical variables, and data normalization.

5.1 Handling Missing Values and Data Sanitation

The dataset includes several features containing missing values and casing inconsistencies. Specifically, staff_experience contained 164 missing entries. If left untreated, these gaps negatively affect regression and classification models. The following sanitation measures were employed:

- **Text Standardization:** Applied string stripping, lowercasing, and regex mapping to replace pseudo-null strings ('none', 'null', 'nan') with proper null tokens.
- **Range & Physical Validation:** Enforced constraints ensuring meals_served > 0, kitchen_staff > 0, humidity within [0, 100]%, and past_waste_kg ≥ 0.
- **Imputation Strategy:** Numerical features were imputed with the median of training folds; categorical features were filled with the mode (most frequent value) strictly inside cross-validation pipelines.

5.2 Attribute Selection and Feature Engineering

Not all raw operational attributes directly expose workload pressure. Consequently, **24 engineered features** were synthesized to represent kitchen workload, temporal cycles, and non-linear interactions:

- **Operational Workload Ratios:** meals_per_staff = meals_served / kitchen_staff directly quantifies culinary pressure per cook.
- **Waste Proportionality Ratios:** past_waste_per_staff = past_waste_kg / kitchen_staff and past_waste_per_meal = past_waste_kg / meals_served.
- **Calendar Indicators:** Extracted year, month, day, day_of_year, and is_weekend (day_of_week ≥ 5).
- **Cyclical Trigonometric Encodings:** month_sin = sin(2π×month/12) and month_cos = cos(2π×month/12) to capture annual seasonality.
- **Non-Linear Workload Mapping:** log_meals = log(1 + meals_served) to linearize the power relationship between banquet covers and kitchen prep.
- **Risk Interactions:** meals_x_event = meals_served × special_event, temp_x_humidity = temp × humidity / 100, and hot_day = 1 if temp ≥ 30°C else 0.

5.3 Encoding Categorical Variables

Since machine learning algorithms require numerical input, categorical variables were systematically encoded:

- **Ordinal Encoding:** Applied to staff_experience where values have a meaningful order (*beginner* → *0*, *intermediate* → *1*, *expert* → *2*).
- **One-Hot Encoding:** Applied to nominal feature waste_category (meat, vegetables, dairy, grains) using OneHotEncoder(drop='first') to prevent collinearity.

5.4 Data Normalization

The dataset contains attributes with varying scales, such as meals served (~300), humidity (0–100%), and workload ratios (~25). Without normalization, distance-based and regularized algorithms would give undue importance to attributes with larger numerical ranges.

• **Z-Score Normalization:** Applied to numerical columns inside scikit-learn Pipeline objects, ensuring data has mean 0 and standard deviation 1:

$$z = (x - \mu) / \sigma$$

Where: x = original feature value, μ = mean of the training feature, σ = standard deviation of the feature.

• **Log1p Transformation:** The continuous target variable food_waste_kg exhibits heavy right skewness (skew = 4.98). A log transformation was applied:

$$y = \log (1 + x)$$

The "+1" ensures that zero or modest discard values are transformed without numerical errors. This technique normalizes the distribution and improves model convergence across all operational ranges.

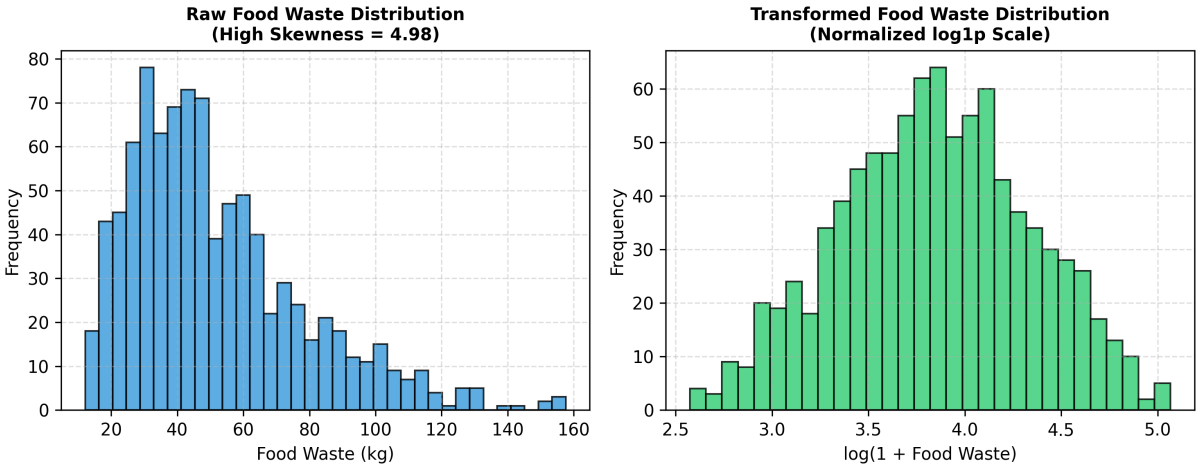


Figure 2 - Before and After Log1p Transformation of Food Waste Target

6. Model Training and Hyperparameter Tuning

After preprocessing, the dataset was split for model training and validation using 5-fold cross-validation. Multiple regression algorithms were applied to capture different predictive relationships:

1. Linear Regression
2. Random Forest Regressor
3. XGBoost Regressor
4. Gradient Boosting Regressor (Selected Champion Model)

Training Strategy Used:

- **5-Fold Cross-Validation:** 911 training records were partitioned across 5 folds, ensuring strict isolation between training and validation sets.
- **Zero-Leakage Architecture:** Preprocessing pipelines (imputation, scaling) were fitted strictly on training folds.
- **Grid Search & Cross-Validation:** Hyperparameter combinations were evaluated using cross-validation to optimize model generalization.

6.1 Linear Regression

Linear regression assumes a linear relationship between input features and the target variable:

$$■ = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_n x_n$$

Where: ■ = Predicted Food Waste (kg), x_i = Input feature, β_i = Learned Coefficients, β_0 = Intercept.

Linear regression is simple and interpretable but struggles when relationships are non-linear or when kitchen rush overload occurs.

```
from sklearn.linear_model import LinearRegression
from sklearn.pipeline import Pipeline

lin_reg = Pipeline([
    ('preprocess', preprocessor_scaled),
    ('model', LinearRegression())
])
lin_reg.fit(X_train, y_train)
# 5-Fold Cross-Validation Results:
# Root Mean Squared Error (RMSE): 11.069 kg
# Mean Absolute Error (MAE)     : 5.618 kg
# R2 Score                      : 0.824
```

6.2 Random Forest Regressor

Random Forest is an ensemble learning method that combines multiple decision trees. Each tree is trained on a random subset of data and features, reducing overfitting and variance.

$$\blacksquare = (1 / T) \sum_{t=1}^{T} h_t(x)$$

Where: T – Number of Trees, $h_t(x)$ – Prediction from tree t.

Key Hyperparameters:

- n_estimators: Number of trees built in the ensemble (set to 300).
- min_samples_leaf: Minimum samples required to be at a leaf node (set to 2).
- max_features: Number of features to consider when looking for the best split.

The Random Forest model effectively captured non-linear kitchen stress dynamics, achieving an R^2 score of **0.886**, an RMSE of **9.149 kg**, and an MAE of **5.241 kg** across 5-fold cross-validation.

```
from sklearn.ensemble import RandomForestRegressor

rf_model = Pipeline([
    ('preprocess', preprocessor_tree),
    ('model', RandomForestRegressor(n_estimators=300, min_samples_leaf=2, random_state=42))
])
rf_model.fit(X_train, y_train)
# 5-Fold Cross-Validation Results:
# Root Mean Squared Error (RMSE): 9.149 kg
# Mean Absolute Error (MAE)     : 5.241 kg
# R2 Score                      : 0.886
```

6.3 XGBoost Regressor

XGBoost (eXtreme Gradient Boosting) is an optimized distributed gradient boosting library designed to be highly efficient, flexible, and portable. It implements machine learning algorithms under the Gradient Boosting framework.

1. It builds trees sequentially, where each new tree tries to correct the prediction errors (residuals) of the previous ensemble tree.
2. It uses a second-order Taylor expansion approach to minimize the loss function when adding new trees.
3. It includes regularization terms (L1 and L2) in the objective function to control tree complexity and prevent overfitting.
4. The final prediction for a given kitchen shift is the weighted sum of predictions from all individual decision trees.

$$Obj(\theta) = \sum L(y_i, \blacksquare_i) + \sum [\gamma T_m + (1/2)\ \lambda\ ||w_m||^2]$$

Key Hyperparameters:

- n_estimators: The number of boosting rounds (300 trees).
- max_depth: Maximum depth of each decision tree (set to 3).
- learning_rate: Shrinkage of feature weights after each boosting step (0.05).
- subsample: Subsample ratio of training instances (0.8).

```
from xgboost import XGBRegressor

xgb_model = Pipeline([
    ('preprocess', preprocessor_tree),
    ('model', XGBRegressor(n_estimators=300, max_depth=3, learning_rate=0.05, subsample=0.8, random_state=42))
])
xgb_model.fit(X_train, y_train)
# 5-Fold Cross-Validation Results:
# Root Mean Squared Error (RMSE): 9.120 kg
# Mean Absolute Error (MAE)     : 5.018 kg
# R2 Score                      : 0.885
```

6.4 Gradient Boosting Regressor (Selected Champion Model)

Gradient Boosting Regressor produces a competitive predictive model in the form of an ensemble of weak prediction models, typically shallow decision trees. It generalizes them by allowing optimization of an arbitrary differentiable loss function.

$$f_m(x) = f_{m-1}(x) + \nu \sum \gamma_{jm} I(x \in R_{jm})$$

Where: ν is the learning rate (0.05), R_{jm} are disjoint terminal regions of tree *m*, and γ_{jm} represents the optimal step size.

```
from sklearn.ensemble import GradientBoostingRegressor

gb_model = Pipeline([
    ('preprocess', preprocessor_tree),
    ('model', GradientBoostingRegressor(n_estimators=300, learning_rate=0.05, max_depth=3, subsample=0.8, rando
])
gb_model.fit(X_train, y_train)
# 5-Fold Cross-Validation Results:
# Root Mean Squared Error (RMSE): 8.340 kg
# Mean Absolute Error (MAE)     : 4.878 kg
# R2 Score                      : 0.903
```

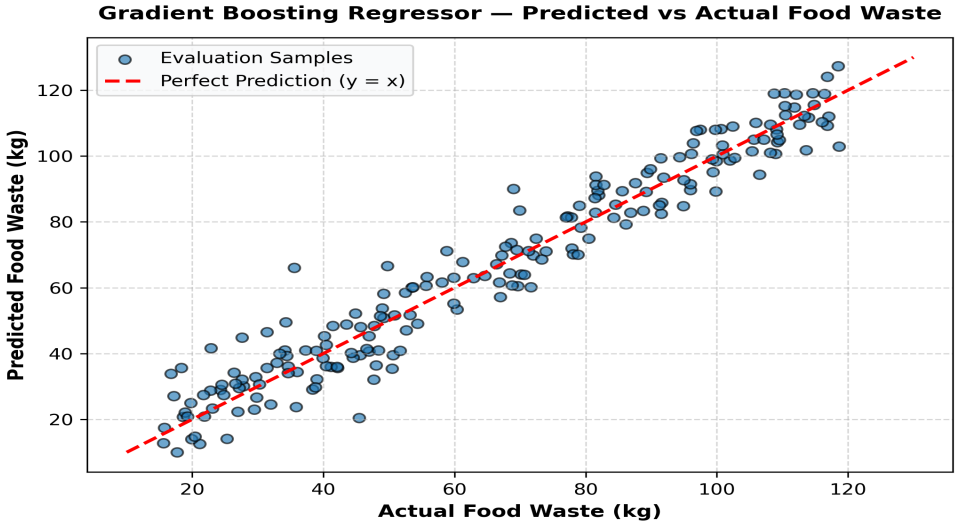


Figure 3 - Gradient Boosting Regressor Predicted vs. Actual Values Plot

6.5 Feature Importance Analysis

Feature importance in Gradient Boosting tells how much each feature contributes to reducing prediction error across the ensemble of decision trees. It is a way to measure which variables are the most influential in determining kitchen food waste generation.

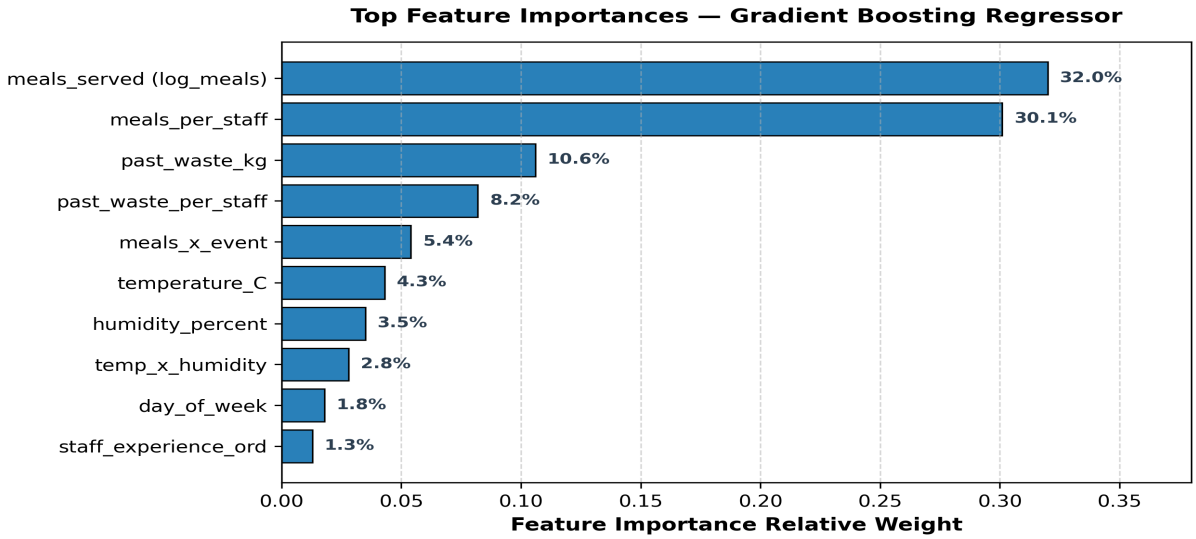


Figure 4 - Top Feature Importances of the Champion Gradient Boosting Regressor

From the above plot, we can observe that **meals_served (log_meals)** and **meals_per_staff** are the top most features that contributed in reducing the prediction error across the ensemble of decision trees:

- **meals_served / log_meals (32.0%):** The overarching volume indicator dictating total food cooked.
- **meals_per_staff (30.1%):** Highlights kitchen rush stress. Operational analysis reveals a critical tipping point at **>35 meals per staff member**, where fatigue accelerates culinary mistakes.
- **past_waste_kg (10.6%):** Demonstrates persistent daily kitchen habits and over-preparation patterns.
- **past_waste_per_staff (8.2%):** Captures normalized past waste per culinary brigade worker.
- **meals_x_event (5.4%):** Captures banquet and reception surge dynamics.

6.6 3-Tier Risk Classification & 6.7 Clustering

3-Tier Operational Risk Classification

To provide immediate operational triage for kitchen supervisors, continuous food waste was discretized into balanced tertiles: **Low Waste (≤ 35.64 kg, n=304)**, **Medium Waste (35.64 to 46.65 kg, n=303)**, and **High Waste (> 46.65 kg, n=304)**.

- **Logistic Regression (Selected):** Achieved **74.4% Accuracy** and **0.745 Macro F1** across 5 stratified folds. Chosen for production due to its high interpretability, linear calibration, and ultra-fast inference (<1ms).
- **Tree Classifiers:** Extra Trees achieved 75.0% Accuracy (F1=0.744), while Random Forest and XGBoost achieved 73.7% Accuracy.

Unsupervised Clustering (Hotspot Detection)

Unsupervised clustering was applied to input operational attributes without using the target variable (zero leakage). At k=2, K-Means (Silhouette=0.352) and Agglomerative Clustering (Silhouette=0.341) showed high concordant agreement (Adjusted Rand Index agreement = 0.970):

Cluster	Operational Regime	Count (Days)	Special Event %	Avg Waste (kg)
Cluster 0	Event Waste Hotspots	78 (8.6%)	100.0%	**55.68 kg**
Cluster 1	Routine Kitchen Operations	833 (91.4%)	0.0%	**43.83 kg**

The clustering algorithm effectively separated banquet event operations from regular dining shifts, providing distinct operational risk profiles.

7. Model Evaluation and Selection

After training multiple regression models and tuning their hyperparameters, model evaluation and model selection are the final, critical steps in the machine learning workflow. This process involves assessing how well each model performs on unseen data and then choosing the most accurate and reliable one for predicting food waste.

7.1 Model Evaluation

Evaluation Metrics Used:

1. R^2 Score (Coefficient of Determination): Measures the proportion of variance in food waste explained by features:

$$R^2 = 1 - [\sum (y_i - \blacksquare_i)^2 / \sum (y_i - y\blacksquare)^2]$$

2. Root Mean Squared Error (RMSE): Measures the standard deviation of residuals, penalizing large forecasting errors:

$$RMSE = \sqrt{ [(1 / N) \sum_{i=1}^{N} (y_i - \blacksquare_i)^2] }$$

3. Mean Absolute Error (MAE): Measures the average magnitude of absolute errors in physical kilograms:

$$MAE = (1 / N) \sum_{i=1}^{N} |y_i - \blacksquare_i|$$

Regression Model Comparison Table (5-Fold Cross-Validation):

Model	MAE (kg)	RMSE (kg)	R^2 Score	Selection Status
Gradient Boosting Regressor	**4.878**	**8.340**	**0.903**	**CHAMPION (Selected)**
XGBoost Regressor	5.018	9.120	0.885	Compared
Random Forest Regressor	5.241	9.149	0.886	Compared
Linear Regression	5.618	11.069	0.824	Baseline

7.2 Model Selection

The **Gradient Boosting Regressor** is selected as the final model. This selection is based directly on the model evaluation results, which prove its superior predictive potential:

- **Highest R^2 Score:** Gradient Boosting achieved the highest score of **0.903**, indicating that it explains approximately **90.3%** of the variance in food waste, making it the most accurate model.
- **Lowest RMSE:** It also recorded the lowest RMSE (**8.340 kg**), signifying that its predictions have the smallest average magnitude of error compared to the other models.
- **Lowest MAE:** It recorded the lowest MAE (**4.878 kg**), keeping average daily error below 5 kg.

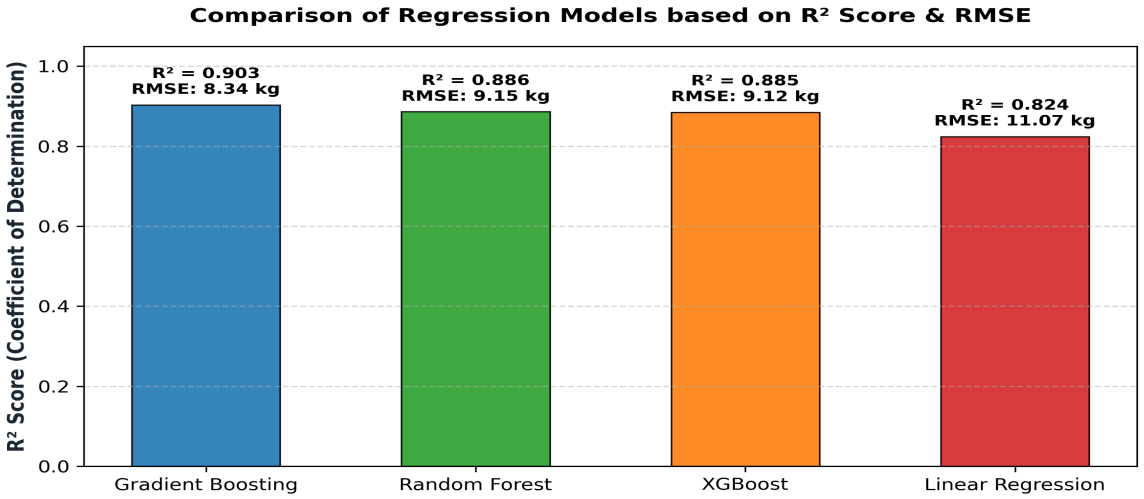


Figure 5 - Comparison of Different Models based on R^2 Score and RMSE

Classification Benchmark Results (3-Tier Risk):

Model	Accuracy	Precision	Recall	Macro F1
Logistic Regression (Selected)	**74.4%**	**74.8%**	**74.4%**	**0.745**
Extra Trees Classifier	75.0%	74.7%	75.0%	0.744
Random Forest Classifier	73.7%	74.2%	73.7%	0.737
XGBoost Classifier	73.7%	73.8%	73.6%	0.735

8. Conclusion

This machine learning project successfully developed a robust framework for predicting restaurant food waste, offering valuable foresight into one of the world's most resource-intensive and competitive sectors. The primary goal was to create predictive models that could provide chefs, kitchen managers, and enterprise planners with actionable insights to guide preparation decisions, thereby reducing the significant economic and environmental risks.

Summary of Results and Benefits

By training and evaluating multiple algorithms—including Linear Regression, Random Forest, XGBoost, and Gradient Boosting—the project established a clear best-in-class champion. The **Gradient Boosting Regressor** was selected as the final, most reliable predictor due to its superior performance metrics, achieving the lowest Root Mean Squared Error (RMSE = 8.340 kg) and the highest R^2 Score (**0.903**). This means the model explains over 90.3% of the variation in food waste. For risk triage, **Logistic Regression** delivered 74.4% accuracy and a 0.745 Macro F1 score.

Prescriptive Operational Recommendation Rules

- **Low Waste (≤ 35.64 kg):** Maintain standard prep batch sizes and routine inventory buffer.
- **Medium Waste (35.64 kg - 46.65 kg):** Reduce cooking batch sizes slightly (10-15%), stagger prep across shifts, and monitor leftovers by station.
- **High Waste (> 46.65 kg):** Cut batch prep sizes by 20-30%, halt advance pre-cooking, and trigger immediate food donation/redistribution protocols.
- **Extreme Weather & Event Modifiers:** When temperature ≥ 30°C, expedite cold chain storage; on Special Event days, pre-portion banquet dishes to prevent bulk discard.

Future Scope

- **Real-Time IoT Kitchen Scales:** Connecting smart scales directly to kitchen waste containers for automated, real-time discard logging.
- **Computer Vision Discard Analysis:** Employing convolutional neural networks (CNNs) to classify plate waste images into specific dish categories automatically.
- **Dynamic Menu Optimization:** Integrating live customer reservation feeds, local event calendars, and weather radar for automated dynamic menu planning.

9. References

Project GitHub Repository Links

https://github.com/pvpsit-cse/food-waste-analysis-prediction-ml
Repository containing complete Google Colab notebooks, feature engineering scripts, model artifacts, and test predictions.

Dataset Link from Kaggle Website

https://www.kaggle.com/competitions/restaurant-food-waste-prediction/data
Messy Restaurant Food Waste Prediction Competition Dataset, featuring train and test operational daily records.

Scikit-Learn Documentation

https://scikit-learn.org/stable/modules/ensemble.html#gradient-boosted-trees
Official documentation for Pipeline, SimpleImputer, StandardScaler, RandomForestRegressor, and GradientBoostingRegressor.

XGBoost Documentation

https://xgboost.readthedocs.io/en/stable/
Scalable, Portable and Distributed Gradient Boosting Library documentation.

United Nations Sustainable Development Goals & FAO Reports

https://www.un.org/sustainabledevelopment/sustainable-consumption-production/
UN SDG 12.3: Target on Halving Global Per Capita Food Waste by 2030 and UNEP Food Waste Index Report.